Anup Raj Niroula

Machine Learning Engineer | MLOps ● Scalable ML Systems ● Cloud-Native AI

a.r.niroula@gmail.com | linkedin.com/in/anup-raj-niroula | github.com/ARNiroula | arniroula.github.io

Education

New York University — New York, USA
Master of Science in Computer Engineering ***– GPA: 4.0/4.0*** — *September 2024 – May 2026*

Kathmandu University — Bagmati, Nepal
Bachelor of Engineering in Computer Engineering ***– GPA: 3.83/4.0*** — *August 2017 – March 2022*

Technical Skills

Languages: Python, Go, R, JavaScript/TypeScript, SQL (Postgres, MySQL, SQLite), C/C++, Bash
Libraries: Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, Tidyverse, data.table
Frameworks: FastAPI, Django, Flask, React, NextJS, Astro, Node.js
Cloud Platforms: Amazon Web Services, Google Cloud Platform, Chameleon Cloud
Developer Tools: Git, Docker, Kubernetes, GitHub Actions, Argo CD, VS Code, Neovim

Projects

Spotify Buddies | *MLflow, FastAPI, PostgreSQL, Redis, Grafana, Airflow, Docker* — GitHub Link

- Trained and deployed a two-stage collaborative filtering recommender (KNN + Bayesian Personalized Ranking) on Chameleon Cloud, serving up to 200,000 predictions/min with offline similarity computation and MLflow/Ray Tune–tracked training.
- Redesigned the deployment pipeline, reducing model server launch time from 300s to 30s while keeping the image under 2 GB
- Implemented real-time monitoring with Grafana and Prometheus to track model performance and detect cold-start scenarios
- Automated retraining and model lifecycle management using Apache Airflow, improving system scalability and maintainability

News Classification (LoRA Fine-Tuning) | *PyTorch, Transformers, Tensorboard* — GitHub Link

- Fine-tuned RoBERTa and DeBERTa models using LoRA on news datasets with HuggingFace Transformers
- Applied domain adaptation techniques to enhance performance on target-specific news data
- Achieved test accuracy of more than 93%
- Secured 1st place in an internal Kaggle-style competition on LoRA-based fine-tuning organized by NYU

Experience

MedLaunch Concepts — Florida, USA
Machine Learning Engineer Intern (Team Lead) — *January 2026 – Present*

- Leading the Data Science team to design a hospital feasibility analysis system using historical CMS data
- Developing predictive models to forecast hospital operational goals, enabling data-driven strategic planning for clients

AI Engineer Intern — *May 2025 – August 2025*

- Built an event-driven Policy Analyzer using AWS S3, Bedrock, and Lambda to automate parallel analysis of hospital policies
- Engineered a scalable, serverless ETL pipeline for document conversion, managing 1,000+ concurrent parses
- Integrated an AI-powered chatbot using RAG and vector databases, reducing manual query response time by 80%

New York University — New York, USA
Graduate Teaching Assistant — *January 2025 – December 2025*

- Facilitated office hours for graduate-level Machine Learning courses, clarifying optimization, model evaluation, and deep learning concepts
- Revised machine learning assignments by integrating real-world datasets, leading to a 20% improvement in student scores

Nimble Clinical Research — New Jersey, USA
Software Engineer — *October 2021 – May 2023*

- Led the R Programming Team of 6 developers and maintained API communication between Django and R microservices
- Engineered a Report Automation System generating 100,000+ rows in under 60 seconds (9x performance increase)
- Devised an automated unit testing framework running test suites in CI/CD, ensuring code quality and early bug detection
- Trained and deployed supervised learning models for automated clinical data mapping, achieving 70%+ accuracy